

Logistics ETA & Dynamic Pricing Analytics

KPI Walkthrough | Databricks + BigQuery + Go + Tableau + Omni

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CONTEXT

The Problem

- A logistics platform needs to predict delivery ETAs and adjust pricing in real time, on the request hot path.
- Engineering and business teams both need visibility into model performance and pricing outcomes, without waiting on data eng for every new report.
- Legacy BI (Tableau) is entrenched with execs; a newer semantic-layer tool (Omni) is being adopted for self-serve analytics during the transition.

DATA SOURCE

Where the Data Comes From

- Simulated trip request events: pickup distance, regional demand index, surge multiplier, predicted vs. actual ETA, conversion outcome.
- Generated to mirror real peak-hour demand and congestion patterns (morning/evening commute spikes).
- 20,000 synthetic trip requests across 5 regions over a 60-day window.

HYPOTHESIS

What We're Testing For

- Whether a correction model trained on regional demand and time-of-day context meaningfully improves ETA accuracy over the raw prediction.
- Whether surge pricing changes correlate with reduced conversion, the tradeoff a pricing engine has to balance against revenue.
- Whether driver/dasher availability or raw demand is the binding constraint on conversion.

ARCHITECTURE

Stack & Why It Fits

Layer	Tool	Why
Feature engineering / ML	Databricks	Streaming ingestion + ETA model training together (medallion architecture)
Serving warehouse	BigQuery	Fast, cheap for aggregated KPI queries
Real-time serving	Go	Request hot path, target p99 < 50ms, matches real dispatch/pricing services
Legacy/exec BI	Tableau	Existing exec-facing dashboard tooling
Self-serve BI	Omni	Semantic layer, no duplicated metric logic across teams

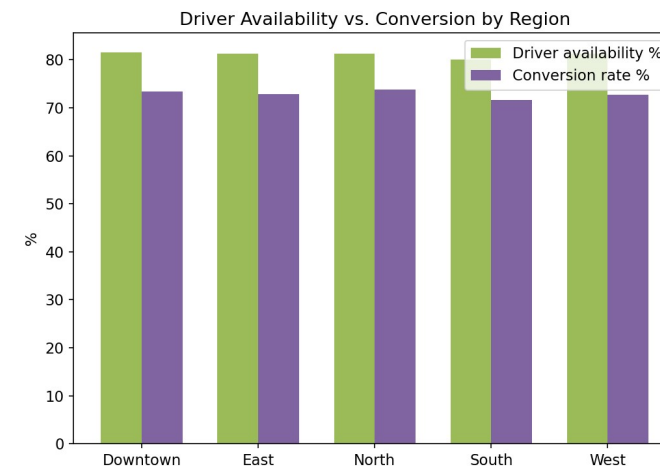
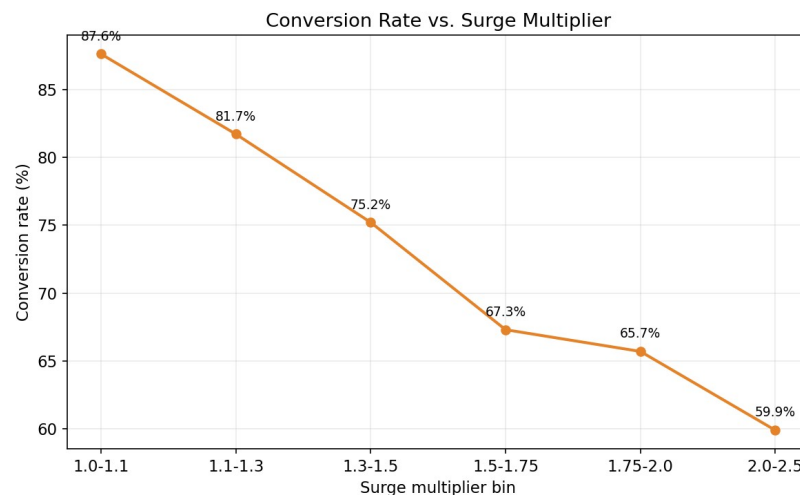
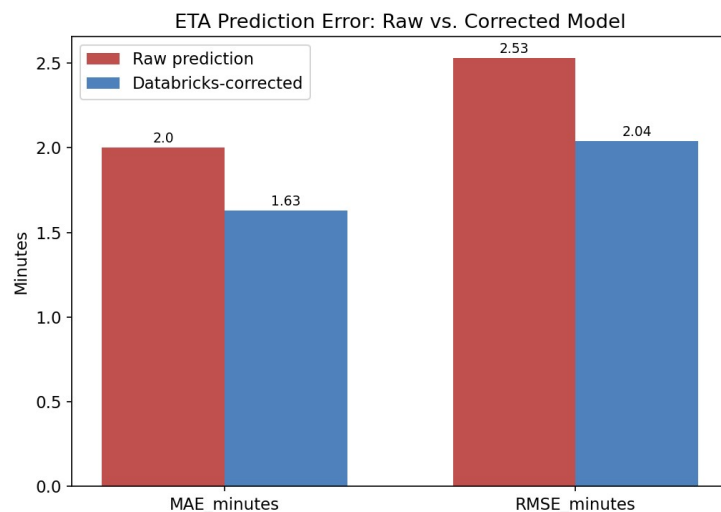
SCOPE

The Task

- Build streaming ingestion and the Bronze/Silver/Gold feature table on Databricks.
- Train an ETA correction model and evaluate it against the raw prediction baseline.
- Implement the Go real-time quote service for ETA + surge pricing.
- Expose gold KPI views in BigQuery to both Tableau and Omni, and validate they return identical numbers.

KPI SNAPSHOT

Results



- ETA MAE improves from 2.00 to 1.63 minutes (RMSE 2.53 to 2.04) with the demand-aware correction model.
- Conversion drops from 87.6% at low surge (1.0-1.1x) to 59.9% at high surge (2.0-2.5x), a clear elasticity signal.
- Driver availability holds steady (~80-82%) across regions and tracks closely with conversion (~72-74%), pointing to utilization as the binding constraint.